Exercise 7
Multistage Amps.

Winter Term 2025/26

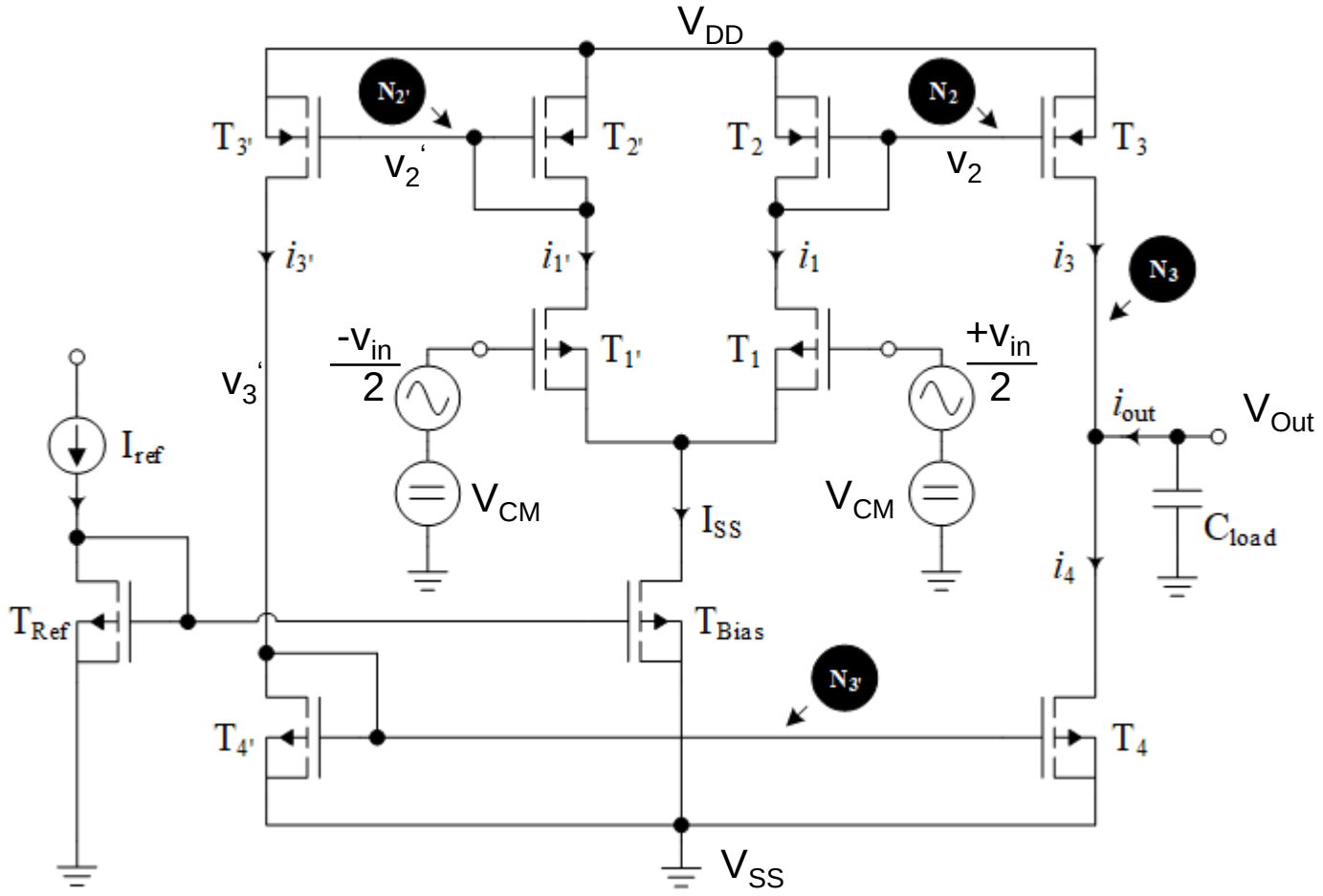
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Exercise 7

Specifications:

Parameter	Value
V_{DD}	1.2V
I_{SS}	100µA
I_{ref}	10µA
V_{ICM}	0.8V
β_n	360µA/V²
β_p	63µA/V²
λ_n, λ_p	0
V_{Tn}	188mV
V_{Tp}	212mV
V_{SB}	0V

All L	10µm
All $U_{GS,eff}$	150mV



Exercise 7-1

1. Dimension the Reference Transistor T_{Ref} and the Tail Current Sink T_{Bias}!

1.1 Current sinked by diode connected Transistor T_{Ref}

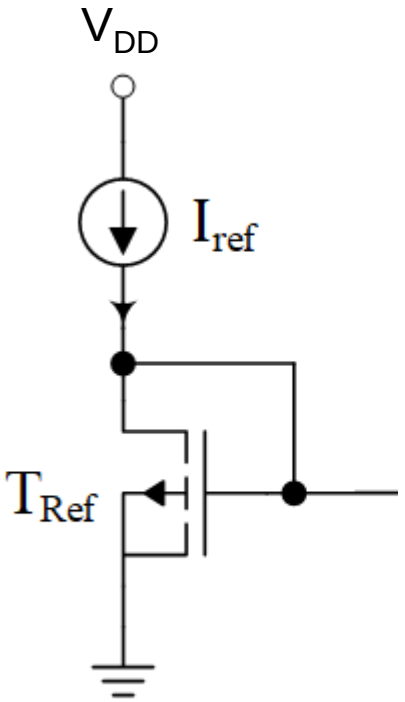
$$I_{Ref} = \frac{\beta_{0n}}{2}\frac{W_{Ref}}{L}V_{GS,eff}^2$$

Solving for W_{Ref}

$$W_{Ref} = 2.47\mu m$$

1.2 Current mirror ratio of 1:10

$$W_{Bias} = 10\, W_{Ref} = 24.7\mu m$$



Exercise 7-2

2. Dimension the differential stage, i.e., the input and the diode-connected load transistors!

Bias current I_{SS} is distributed equally between both inputs, therefore

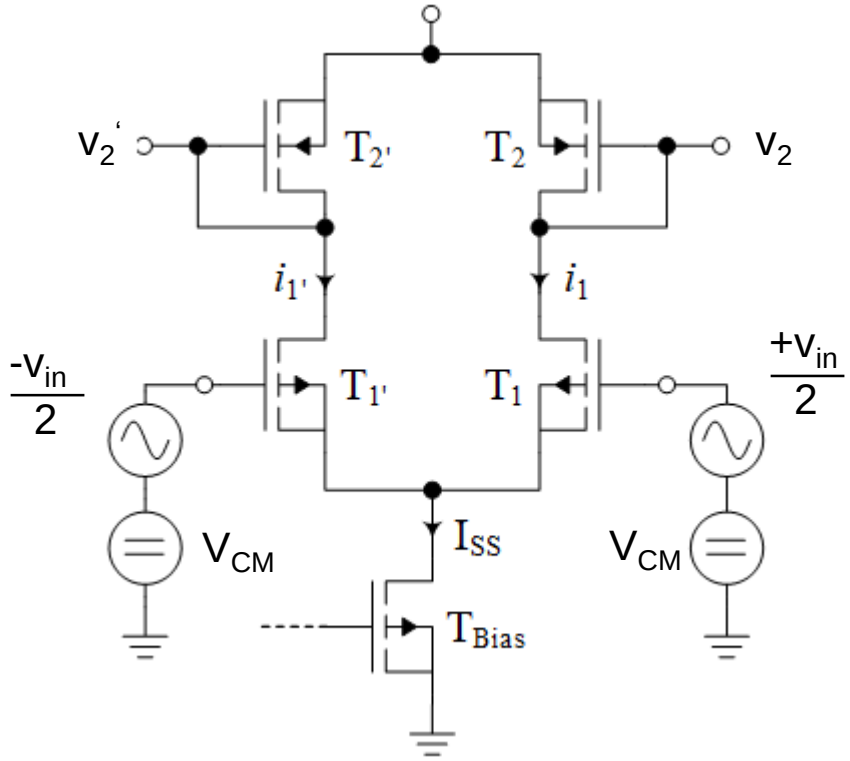
$$I_1 = I_{1'} = \frac{I_{SS}}{2}$$

Since $U_{GS,eff}$ is common for all Transistors

$$W_1 = W_{1'} = \frac{W_0}{2} = 12.35\mu m$$

1.2 The load Transistors T_2 and $T_{2'}$ are diode-connected

$$W_{2,2'} = \frac{I_{SS}}{\beta_{0p}} \frac{L}{V_{GS,eff}^2} = 70.5\mu m$$



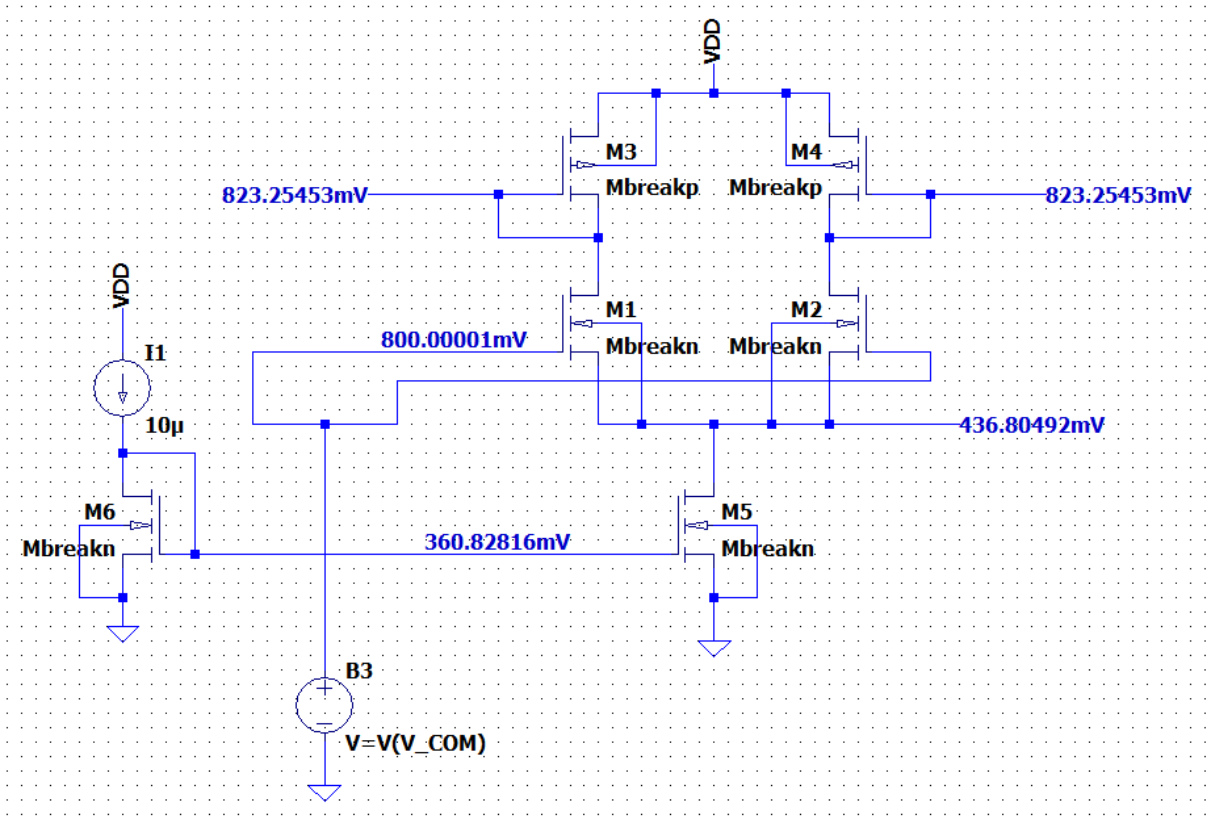
Exercise 7-3

3. Simulate the DC behavior of the differential stage and compare it with your calculations!

$$V_{GS,TRef} = \sqrt{\frac{2I_{Ref}}{\beta_{TRef}}} + V_{T,n} = 338\text{mV}$$

$$V_{N1} = V_{ICM} - V_{GS1} = V_{ICM} - \left(\sqrt{\frac{\frac{2I_{ss}}{2}}{\beta_{T1}}} + V_{Tn}\right) = 462\text{mV}$$

$$V_{Out,2} = V_{DD} - |V_{GS2}| = V_{DD} - \left(\sqrt{\frac{\frac{2I_{ss}}{2}}{\beta_{T2}}} + V_{Tn}\right) = 838\text{mV}$$



Exercise 7-4

4. Dimension the transistors of the (outer) common-source stages (all current mirrors ratios: 1:1)!

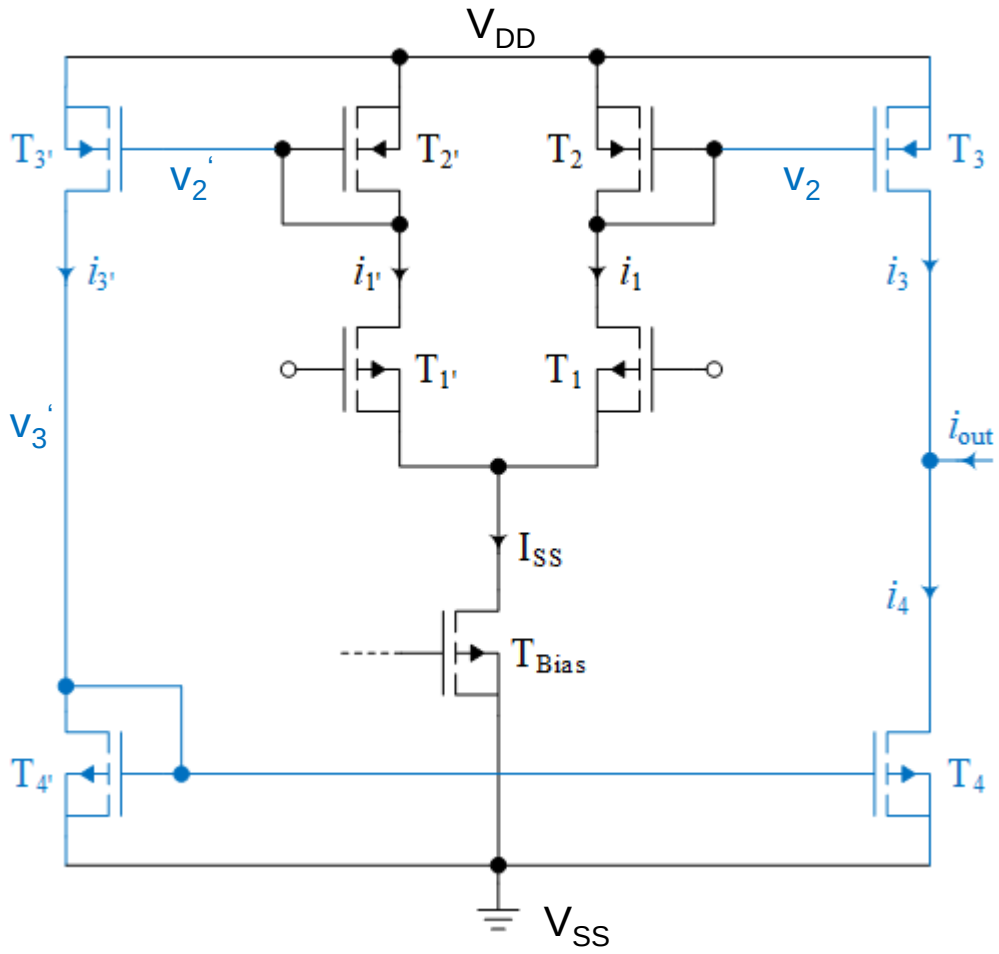
Mirror Ratio of 1:1 for Transistors T_3 and T_2

$$W_2 = W_3 = 70.5\mu m$$

Same applies to Transistors T_3, and T_2,.

A Ratio of 1:1 is assumed for T_4 and T_4'

$$W_4 = \frac{I_{SS}}{\beta_{0n}} \frac{L}{V_{GS,eff}^2} = 12.35\mu m$$

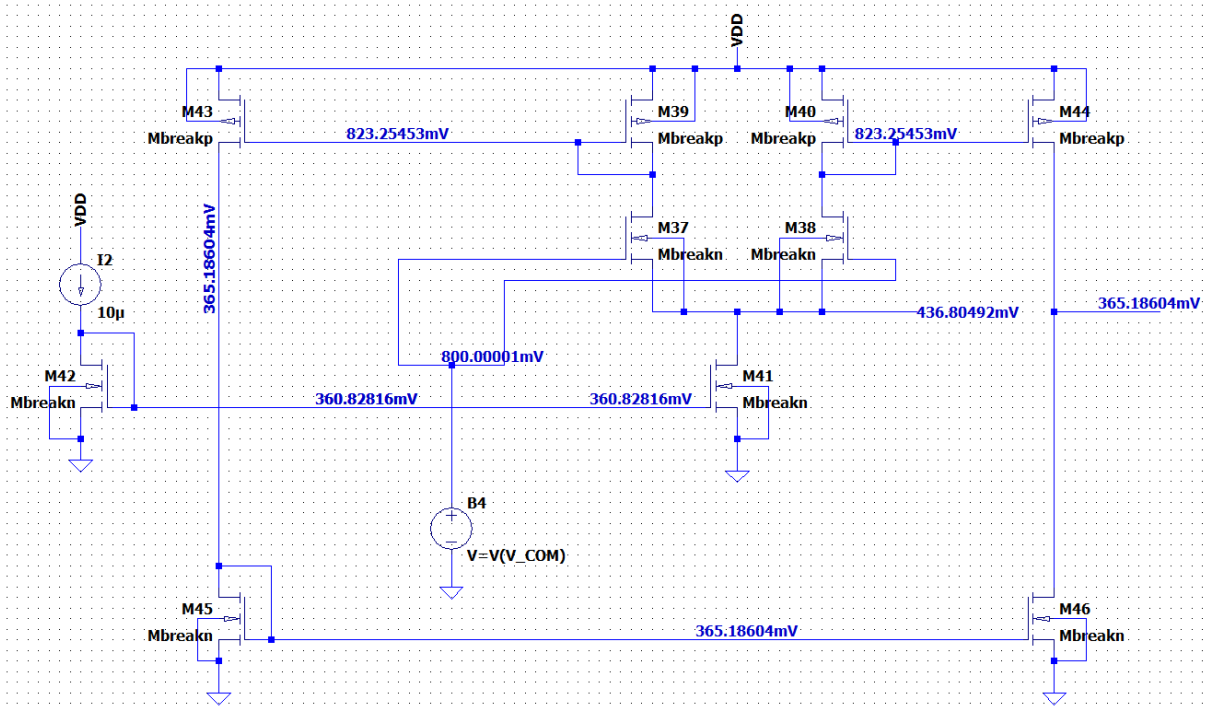


Exercise 7-5

5. Simulate the DC behavior of the single-stage amplifier. Do you consider it critical that the output common mode is not at $V_{DD}/2$?

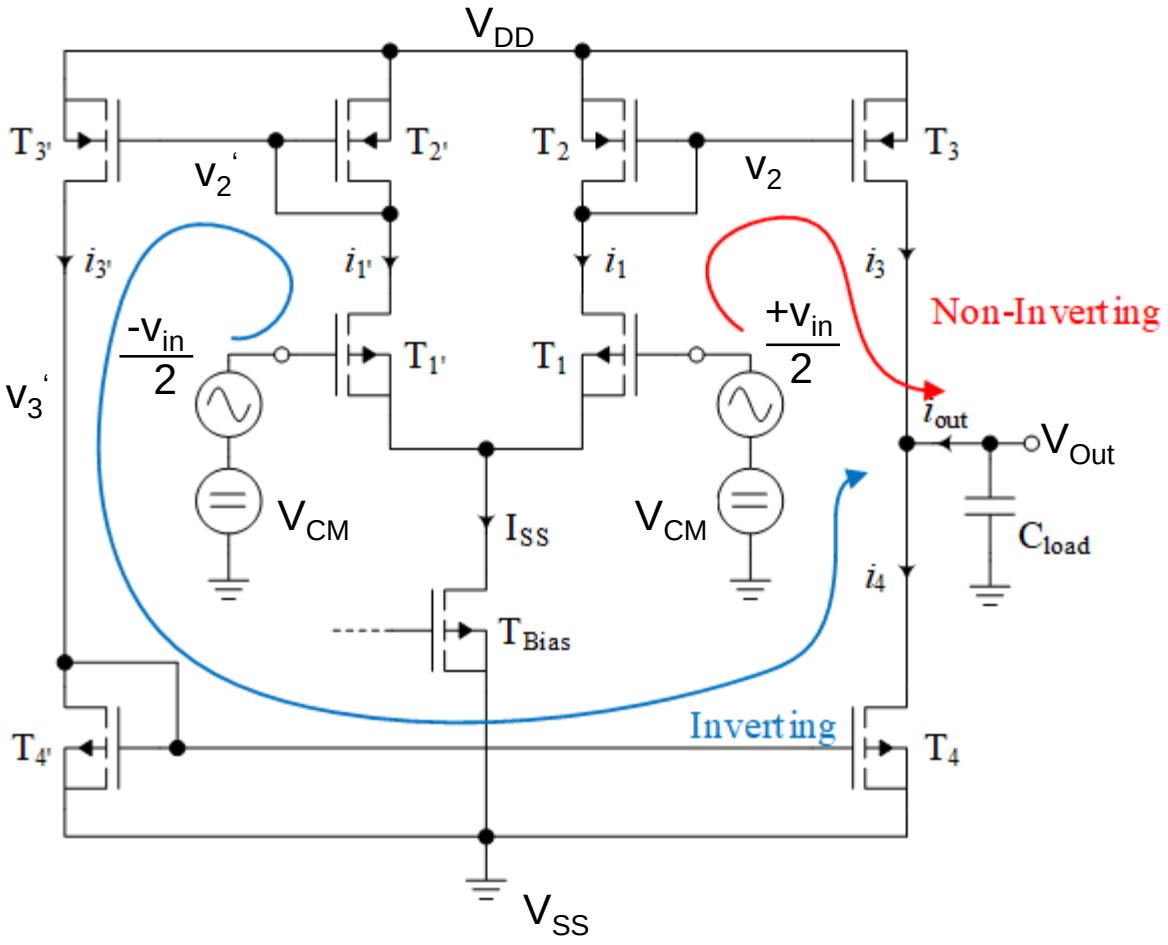
As long as the output transistors are operating in saturation, it is not considered critical if the output common-mode voltage does not result in $U_{DD}/2$. However, the small-signal gain cannot correctly be determined by means of simulations if the transistors are operating in the linear region or even turned off.

The non-centered output prevents the amplifier from using its complete output swing when being used in an open-loop configuration. Differential stages are thus typically used in feedback (closed-loop) configurations.



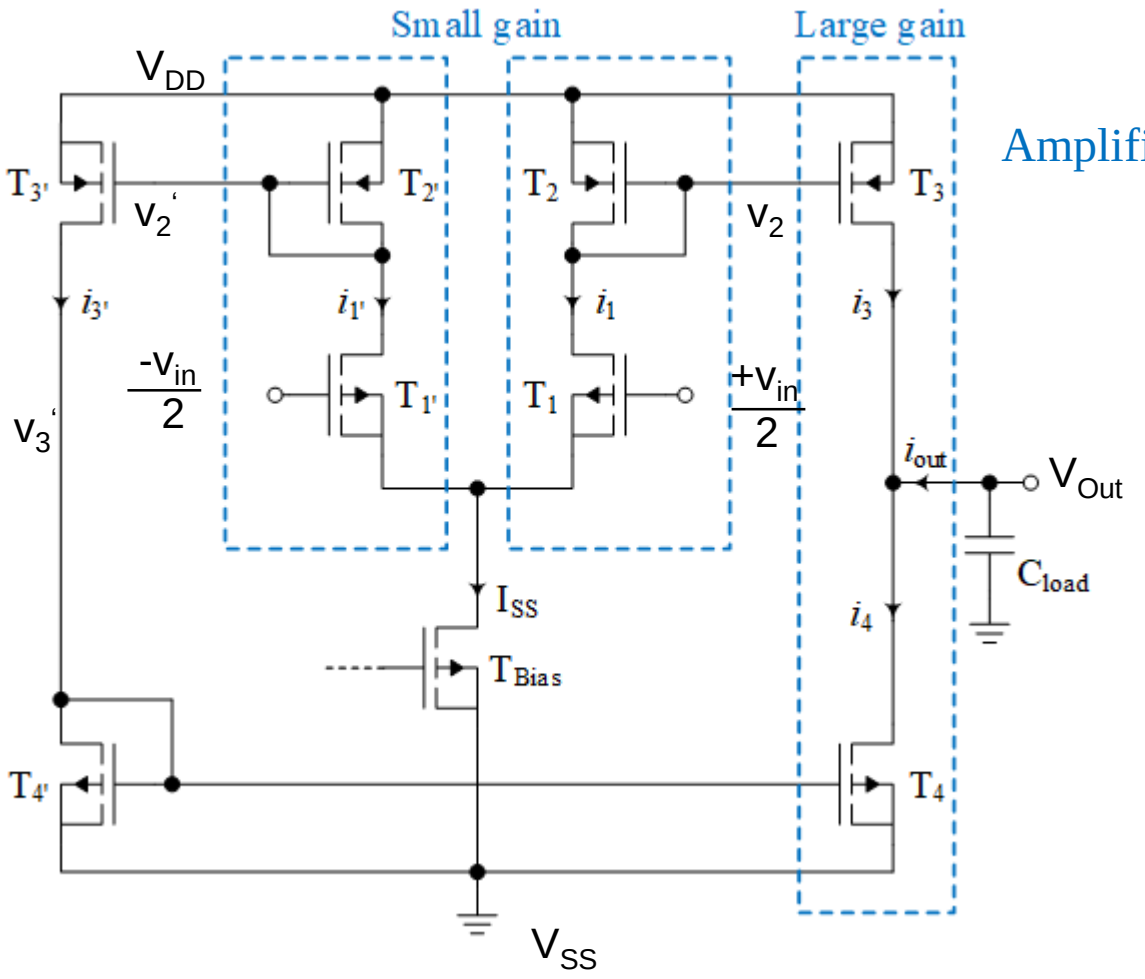
Exercise 7-6

6. Identify the signal paths from the inputs to the output and the different circuit blocks, i.e., amplifying stages and current mirrors, in Figure 1!



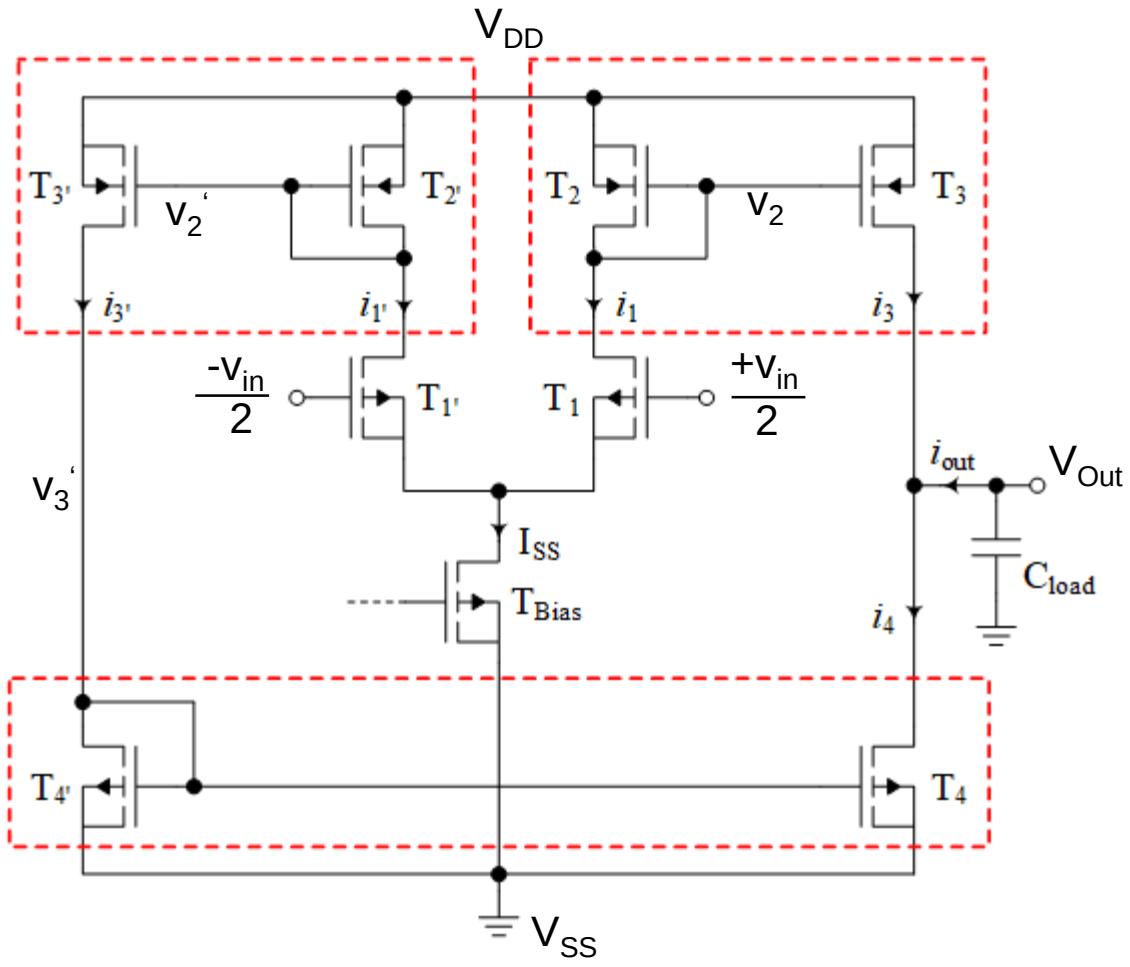
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Exercise 7-6

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Current Mirrors

Exercise 7-7

7. Derive an equation for the differential input to single-ended output gain A_{DM}!

Gain from input to Node 2:

Transistor converting input voltage to current T_1, therefore $g_m = g_{m1}$.

Output resistance $r_{out} = r_{DS1} || \frac{1}{g_{m2}} \approx \frac{1}{g_{m2}}$

Overall Gain $A_{in,2} = \frac{v_2}{v_{in,p}} = -\frac{g_{m1}}{g_{m2}}$

Gain from Node 2 to output:

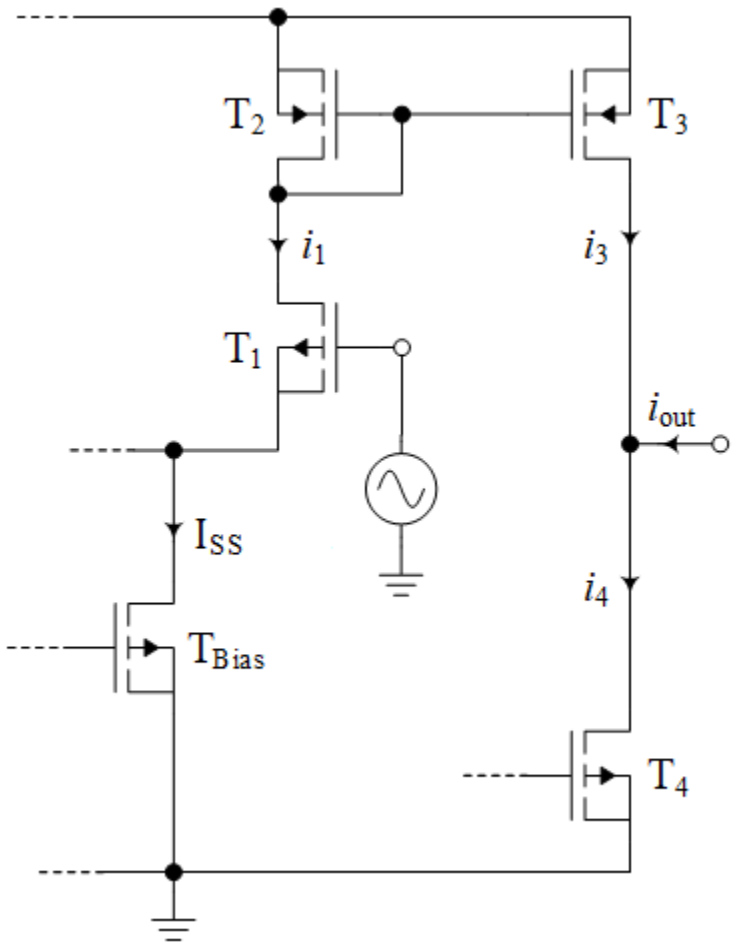
Transistor converting input voltage to current T_3, therefore $g_m = g_{m3}$.

Output resistance $r_{out} = r_{DS3} || r_{DS4}$.

Overall Gain $A_{2,3} = \frac{v_3}{v_2} = -g_{m3} r_{DS3} || r_{DS4}$

Overall Gain of the non-Inverting Path:

$$A_p = \frac{v_{out}}{v_{in,p}} = \frac{g_{m1}}{g_{m2}} g_{m3} r_{DS3} || r_{DS4}$$



Exercise 6-3 – DA-2

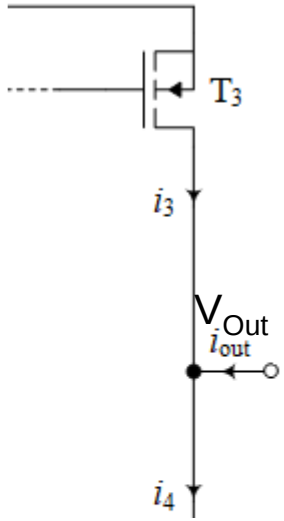
7. Derive an equation for the differential input to single-ended output gain A_{DM}!

Gain from input to Node 2':

Transistor converting input voltage to current $T_{1'}$, therefore $g_m = g_{m1'}$.

Output resistance $r_{out} = r_{DS1'} || \frac{1}{g_{m2'}} \approx \frac{1}{g_{m2'}}$.

Overall Gain $A_{in,2'} = \frac{v_{2'}}{v_{in,n}} = -\frac{g_{m1'}}{g_{m2'}}$



Gain from Node 2' to Node 3':

Transistor converting input voltage to current $T_{3'}$, therefore $g_m = g_{m3'}$.

Output resistance $r_{out} = r_{DS3'} || \frac{1}{g_{m4'}} \approx \frac{1}{g_{m4'}}$.

Overall Gain $A_{2',3'} = \frac{v_{3'}}{v_{2'}} = -\frac{g_{m3'}}{g_{m4'}}$

Gain from Node 3' to output:

Transistor converting input voltage to current T_4, therefore $g_m = g_{m4}$.

Output resistance $r_{out} = r_{DS3} || r_{DS4}$.

Overall Gain $A_{2,3} = \frac{v_3}{v_2} = -g_{m4} r_{DS3} || r_{DS4}$

Overall Gain of the non-Inverting Path:

$$A_p = \frac{v_{out}}{v_{in,n}} = -\frac{g_{m1'}}{g_{m2'}} \frac{g_{m3'}}{g_{m4'}} g_{m4} r_{DS3} || r_{DS4}$$

Exercise 7-7

7. Derive an equation for the differential input to single-ended output gain A_{DM}!

The differential gain of the amplifier is derived by applying superposition, i.e.

$$v_{out} = \frac{1}{2} v_{in} A_p + \left(-\frac{1}{2} v_{in} A_n \right)$$

$$= \frac{1}{2} v_{in} \left(\frac{g_{m1}}{g_{m2}} g_{m3} r_{DS3} || r_{DS4} + \frac{g_{m1'}}{g_{m2'}} \frac{g_{m3'}}{g_{m4'}} g_{m4} r_{DS3} || r_{DS4} \right)$$

This is not yet the equation for the gain, it's just an expression for the output voltage. The gain is equated to

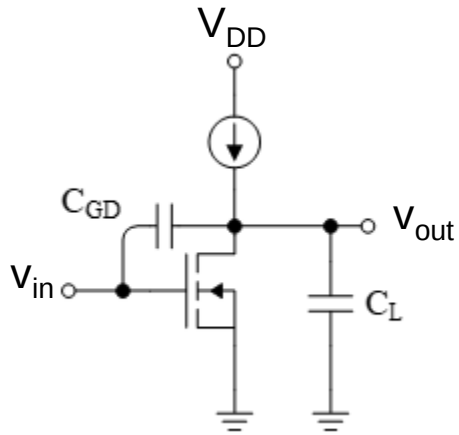
$$A_{DM} = \frac{v_{out}}{v_{in}} = \frac{1}{2} \left(\frac{g_{m1}}{g_{m2}} g_{m3} r_{DS3} || r_{DS4} + \frac{g_{m1'}}{g_{m2'}} \frac{g_{m3'}}{g_{m4'}} g_{m4} r_{DS3} || r_{DS4} \right)$$

Exercise 7-8

8. Determine the number of poles and zeros in each signal path! Analyze the capacitances at every node and give a simplified expression for every pole and zero!

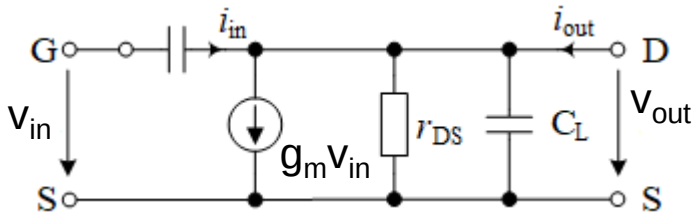
Short Recap:

- Occurrence of Poles: Poles are mostly a consequence of a complex load impedance, e.g. a capacitance in parallel with a resistor.
- Occurrence of Zeros: Zeros appear whenever there is a frequency dependent connection between the input and the output, e.g. the Gate-Drain capacitance in a Common Source Stage.



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The output impedance is the parallel combination of r_{DS}, the load capacitance C_L, as well as the contribution of the Gate-Drain Capacitance (*remember Miller Effect*).

$$z_{out} = \frac{v_{out}}{i_{out}}\Big|_{v_{in}=0} = \frac{r_{DS}}{1 + s\, r_{DS}(C_L + C_{GD})}$$

The frequency dependent g_m is calculated to

$$g_m = \frac{i_{out}}{v_{in}}|_{v_{out}=0} = g_m - sC_{GD}$$

This frequency dependence is introducing a zero in the transfer function

$$A_V = -g_m z_{out} = \frac{(g_m - sC_{GD})r_{DS}}{1 + s\, r_{DS}(C_L + C_{GD})}$$

Exercise 7-8

8. Determine the number of poles and zeros in each signal path! Analyze the capacitances at every node and give a simplified expression for every pole and zero!

The transfer-function might be written in pole-zero form as

$$A_V = A_{DC}\frac{1 - s/z}{1 - s/p}$$

By introducing the **negative pole**

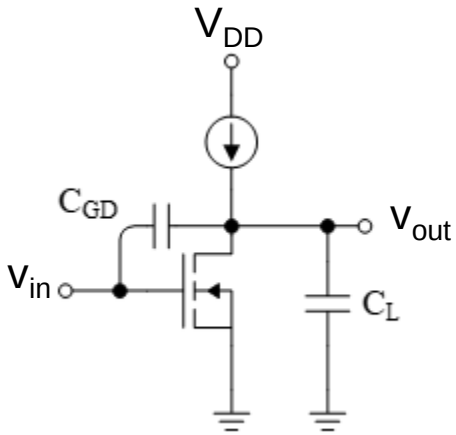
$$p = -\frac{1}{r_{DS}(C_L + C_{GD})}$$

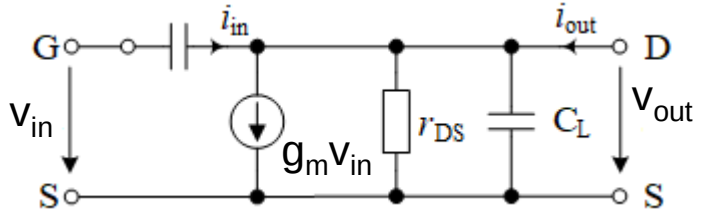
the **positive zero**

$$z = \frac{g_m}{C_{GD}}$$

and the DC gain

$$A_{DC} = g_m r_{DS}$$





Exercise 7-8

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Node 2:

$$C_{GD1} + C_{DB1} + C_{DB2} + C_{GS2} + C_{GS3} + (1 + \boldsymbol{g_{m3} r_{DS3} || r_{DS4}}) C_{GD3}$$

Node 2':

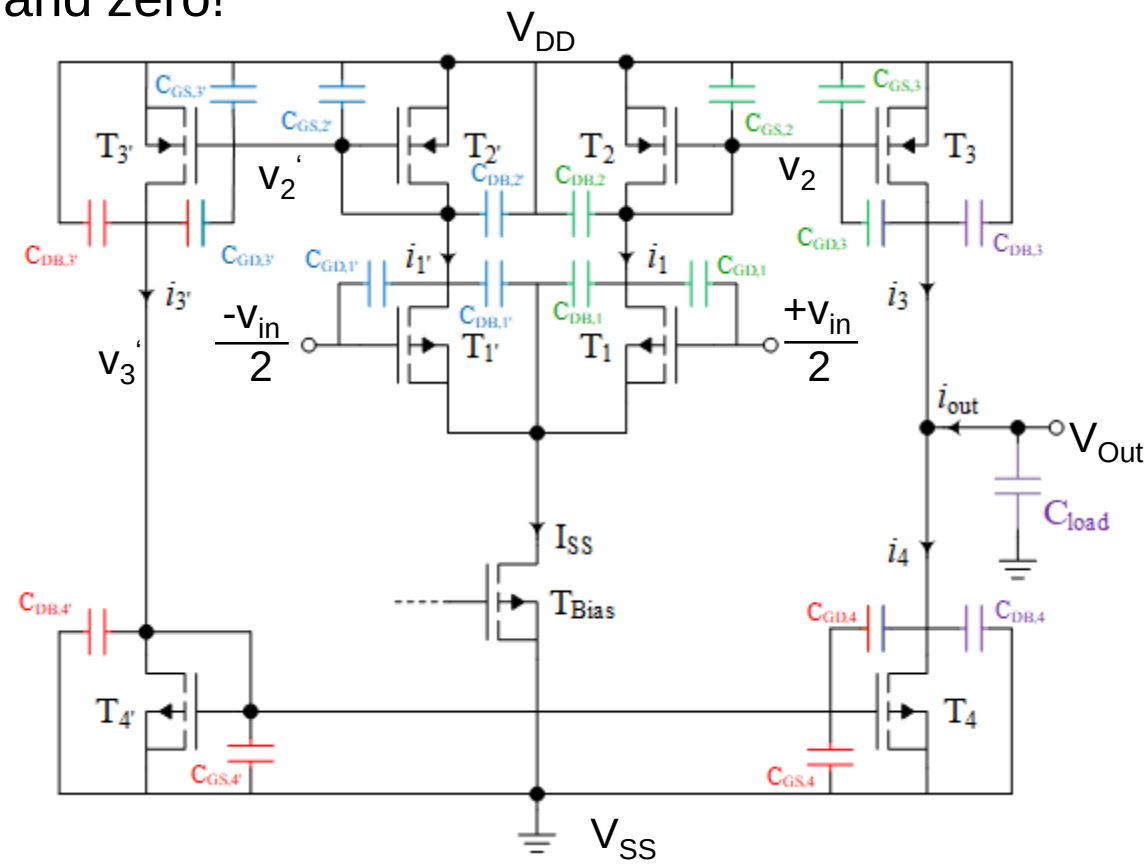
$$C_{GD1'} + C_{DB1'} + C_{DB2'} + C_{GS2'} + C_{GS3'} + (1 + \frac{g_{m3'}}{g_{m4'}}) C_{GD3'}$$

Node 3:

$$C_{GD3} + C_{DB3} + C_{GD4} + C_{DB4} + C_L$$

Node 3':

$$C_{GD3'} + C_{DB3'} + C_{GS4'} + C_{DB4'} + C_{GS4} + (1 + \boldsymbol{g_{m3} r_{DS3} || r_{DS4}}) C_{GD4}$$



Exercise 7-8

8. Determine the number of poles and zeros in each signal path! Analyze the capacitances at every node and give a simplified expression for every pole and zero!

Inverting:

Pole due to Node 2':

$$p_{2'} = -\frac{g_{m2'}}{C_{N2'}} \approx -\frac{g_{m2'}}{C_{GS2} + C_{GS3}}$$

Pole due to Node 3':

$$p_{3'} = -\frac{g_{m4'}}{C_{N3'}} \approx \frac{g_{m4'}}{C_{GS4'} + C_{GS4} + g_{m4'} r_{DS3} || r_{DS4} C_{GD4}}$$

Non-Inverting:

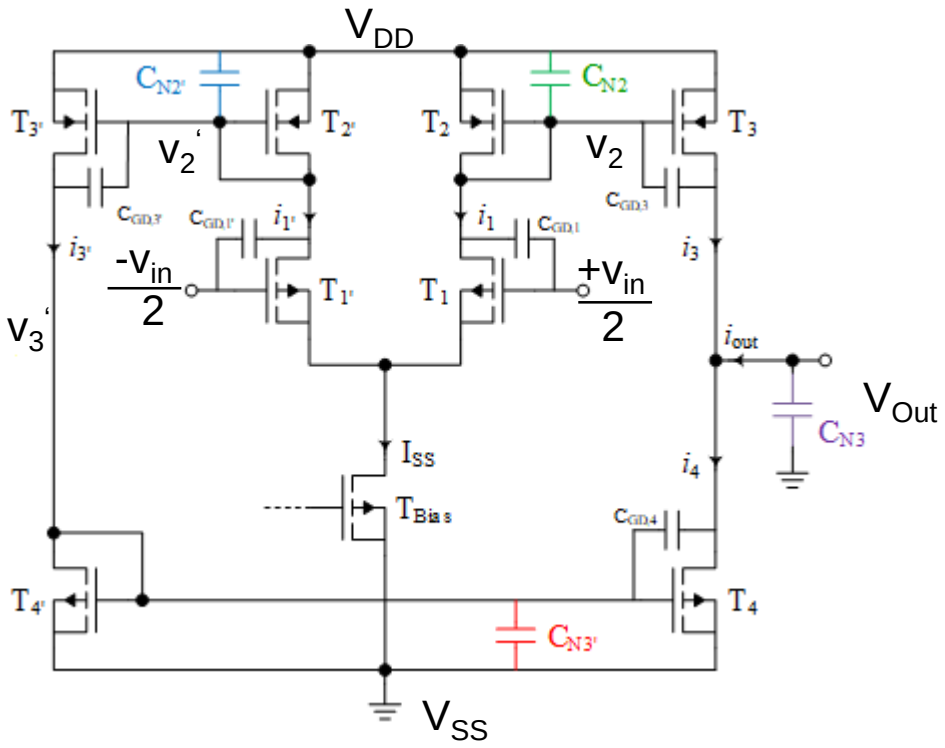
Pole due to Node 2:

$$p_{2'} = -\frac{g_{m2}}{C_{N2}} = -\frac{g_{m2}}{C_{GS2} + C_{GS3} + g_{m3} r_{DS3} || r_{DS4} C_{GD3}}$$

Common:

Pole due to Node 3:

$$p_{out} = -\frac{1}{r_{DS3} || r_{DS4} C_{N3}} \approx -\frac{1}{r_{DS3} || r_{DS4} C_L}$$



Exercise 7-8

8. Determine the number of poles and zeros in each signal path! Analyze the capacitances at every node and give a simplified expression for every pole and zero!

Inverting:

Zero due to $C_{GD1'}$: $z_{1'} = \frac{g_{m1'}}{C_{GD1'}}$

Zero due to $C_{GD3'}$: $z_{3'} = \frac{g_{m3'}}{C_{GD3'}}$

Zero due to C_{GD4}: $z_4 = \frac{g_{m4}}{C_{GD4}}$

Non-Inverting:

Zero due to C_{GD1}: $z_1 = \frac{g_{m1}}{C_{GD1}}$

Zero due to C_{GD3}: $z_3 = \frac{g_{m3}}{C_{GD3}}$

